INCH-POUND

MS21289D
13 February 2024
SUPERSEDING
MS21289C
15 September 1981

DETAIL SPECIFICATION SHEET

BOLT, MACHINE-STEEL DRILLED DOUBLE HEXAGON EXTENDED WASHER, .250-28 UNFJ-3A

Reinstated as inactive for new design after 13 February 2024 and may be used for acquisition.

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and procurement specification SAE AS7478.

Comments, suggestions, or questions on this document should be addressed to DLA Aviation VEB, 6090 Strathmore Road, Richmond, VA 23237 or emailed to STDZNMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 5306

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T05:07Z
Check the source to verify that this is the current version before use.

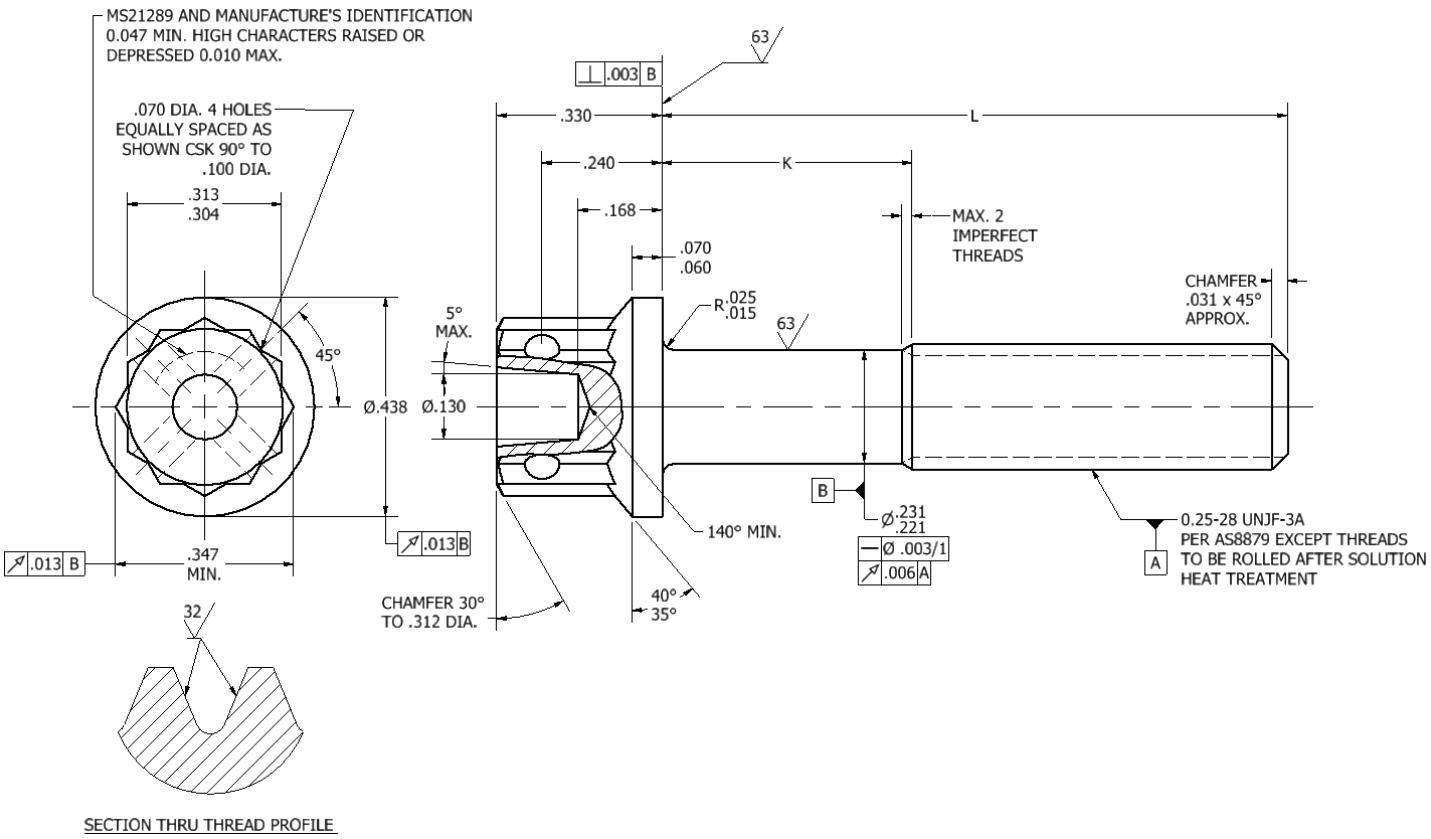


FIGURE I. BOLT

TABLE I. Dash numbers and dimensions

DASH NUMBER	L	K +.000 -.060	APPROX WEIGHT LBS/100
-05	.438	1/	1.22
-06	.500	1/	1.29
-07	.562	1/	1.36
-08	.625	1/	1.44
-09	.688	1/	1.51
-10	.750	1/	1.58
-11	.812	1/	1.65
-12	.875	.125	1.72
-13	.938	.188	1.79
-14	1.000	.250	1.86
-15	1.062	.312	1.93
-16	1.125	.375	2.00
-17	1.188	.438	2.08
-18	1.250	.500	2.15
-19	1.312	.562	2.22
-20	1.375	.625	2.29
-21	1.438	.688	2.36
-22	1.500	.750	2.43
-23	1.562	.812	2.50
-24	1.625	.875	2.57
-25	1.688	.938	2.64
-26	1.750	1.000	2.72
-27	1.875	1.125	2.86
-28	2.000	1.250	3.00
-29	2.125	1.375	3.14
-30	2.250	1.500	3.29
-31	2.375	1.625	3.43
-32	2.500	1.750	3.57
-33	2.625	1.875	3.71
-34	2.750	2.000	3.85
-35	2.875	2.125	4.00
-36	3.000	2.250	4.14
-37	3.125	2.375	4.29
-38	3.250	2.500	4.43
-39	3.375	2.625	4.58
-40	3.500	2.750	4.72
-41	3.625	2.875	4.85
-42	3.750	3.000	4.99
-43	3.875	3.125	5.13
-44	4.000	3.250	5.27
-45	4.125	3.375	5.42
-46	4.250	3.500	5.56
-47	4.375	3.625	5.71
-48	4.500	3.750	5.85
-49	4.625	3.875	6.00
-50	4.750	4.000	6.14
-51	4.875	4.125	6.29
-52	5.000	4.250	6.43
-53	5.125	4.375	6.56
-54	5.250	4.500	6.70
-55	5.375	4.625	6.84
-56	5.500	4.750	6.99
-57	5.625	4.875	7.13
-58	5.750	5.000	7.28
-59	5.875	5.125	7.42
-60	6.000	5.250	7.57

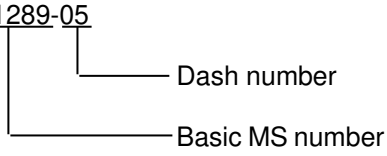
1/ Thread to head. Maximum two incomplete threads. Incomplete threads not to enter fillet radius.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T05:07Z
Check the source to verify that this is the current version before use.

Requirements:

1. Material: Corrosion and heat resistant steel per AMS5732 (UNS S66286).
2. Threads: Threads in accordance with AS8879. Incomplete threads not to enter fillet radius.
3. Dimensioning and tolerancing: Dimensioning and tolerancing shall be in accordance with ASME Y14.5.
4. Surface texture: shall be in accordance with ASME B46.1. Unless otherwise specified, surface to be 125 microinches.
5. Fluorescent penetrant inspection: Fluorescent penetrant inspection shall be in accordance ASTM-E1417/E1417M.
6. Tolerance: Unless otherwise specified, tolerances shall be for linear dimensions ±.010, angular dimensions ±5°.
7. Edges: Break sharp edges .003 - .015, unless otherwise specified.
8. Part number: The part number shall consist of the basic MS number followed by a dash number from Table I.

Example: MS21289-05



MS21289-05 indicates: Bolt, Machine, Steel Drilled Double Hexagon Extended Washer; .250-28 UNJF-3A; Length .438.

Notes:

1. All dimensions are in inches.
2. Do not use unassigned part numbers.
3. In the event of a conflict between the text of this standard and the references cited herein, the text of this standard shall take precedence.
4. Referenced Government (or nongovernment) documents of the issue listed in that issue of the department of defense index of specifications and standards (DoDISS) specified in the solicitation form a part of this standard to the extent specified herein.
5. AS and AMS are SAE publications. ASME is an American Society of Mechanical Engineers publication. ASTM is an ASTM international publication.
6. Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T05:07Z
Check the source to verify that this is the current version before use.

APPENDIX A

3D Model for reference only.
Viewing this model requires a PDF viewer that supports 3D models.
Adobe Acrobat is known to support 3D models but there may be other suitable viewers.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T05:07Z
Check the source to verify that this is the current version before use.

Concluding Material:

Custodians:
Navy – AS
Air Force – 11
DLA - GS

Preparing activity:
DLA – GS7

(Project 5306-2024-006)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST database at https://assist.dla.mil/.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T05:07Z
Check the source to verify that this is the current version before use.